Math 1610, Honors Calculus, Fall 2026, Worksheet 5

October 9, 2026

Name _____

Please write only your own name in the space above.

1. Let $f(x) = x^2$. Using the definition of the derivative,

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h},$$

show that $f'(c) = 2c$ for every real number c.